

2026
MATH CP
PAPER Paper 1

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Candidate Number

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HKDSE Mathematics Compulsory Part — Paper 1 (Mock)
PAPER Paper 1
Question-Answer Book

135 minutes (2.25 hours)

Total marks : 105

INSTRUCTIONS

This paper consists of THREE sections, A(1), A(2) and B. Unless otherwise specified, all working must be clearly shown. Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures. The diagrams in this paper are not necessarily drawn to scale.

1. Simplify $\frac{u^9v^{-4}}{(u^{-3}v)^2}$ and express your answer with positive indices.

(3 marks)

- (3 marks)

- (4 marks)

5.

An air purifier is sold at a discount of 30% on its marked price. After selling the purifier, the profit is \$224 and the percentage profit is 32% . Find the marked price of the purifier.

(4 marks)

6.

- (a) Consider the compound inequality $4(x + 2) \geq x - 1$ and $7 - 2x > x - 8$. Solve the compound inequality.
- (b) Write down the least integer satisfying the compound inequality.

(4 marks)

7. (a) The coordinates of the points A and B are (6, -8) and (-3, 5) respectively. A' is the reflection image of A with respect to the line $y = x$. B is rotated clockwise about the origin O through 90° to B'. Write down the coordinates of A' and B'.
- (b) Find the slope of A'B'.

8.

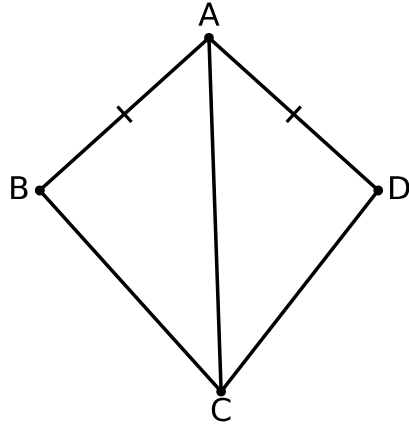


Figure 1

- (a) In Figure 1, ABCD is a quadrilateral. It is given that $AB = AD$ and $\angle BAC = \angle DAC$. Prove that $\triangle ABC \cong \triangle ADC$.
- (b) If $\angle ABC = 100^\circ$ and $\angle BCD = 96^\circ$, find $\angle BAD$.

(5 marks)

- (5 marks)

Section A(2) (35 marks)

10. (a) It is given that $f(x)$ is partly constant and partly varies as x^2 . Suppose that $f(1) = 9$ and $f(3) = 41$. Find $f(x)$.
- (b) Write down the y-intercept of the graph of $y = f(x)$.
- (c) Let k be a real constant. Find the range of values of k such that the equation $f(x) = k$ has two distinct real roots.

(7 marks)

Stem (tens)	Leaf (units)
1	3 6 9
2	1 4 a
3	0 5 b
4	2 8

- The stem-and-leaf diagram in Figure 2 shows the distribution of the ages of the 11 members of a society. It is given that the median and the inter-quartile range of the distribution are 26 and 17 respectively. Find a and b .
- Find the mean and the standard deviation of the distribution.
- A member aged 48 now leaves the society. Find the change in the median of the distribution.

(7 marks)

- (7 marks)

- (7 marks)

- (7 marks)

Section B (35 marks)

15. (a) In a box, there are 7 red balls, 5 blue balls and 3 green balls. 4 balls are randomly drawn from the box at the same time. Find the probability that exactly 2 red balls are drawn.
- (b) Find the probability that at least 1 red ball and at least 1 green ball are drawn.

(6 marks)

16. (a) Let $f(x) = 2x^2 - 8kx + 10k^2 + 24k + 7$, where k is a real constant. Using the method of completing the square, express, in terms of k , the coordinates of the vertex of the graph of $y = f(x)$.
- (b) Find the least value of the y -coordinate of the vertex and the corresponding value of k .
- (c) Someone claims that for any real constant k , the vertex of the graph of $y = f(x)$ lies on the curve $\Gamma : y = \frac{1}{2}x^2 + 12x + 7$. Do you agree? Explain your answer.

(6 marks)

17. (a) Let $A(n)$ be the n th term of an arithmetic sequence. It is given that $A(2) = 11$ and $A(7) = 41$. Find $A(n)$.
- (b) Find $A(1) + A(2) + \dots + A(40)$.
- (c) Let $G(n)$ be the n th term of a geometric sequence with first term $A(3)$ and common ratio 2. Find the least value of n such that $G(1) + G(2) + \dots + G(n) > A(1) + A(2) + \dots + A(40)$.

(6 marks)

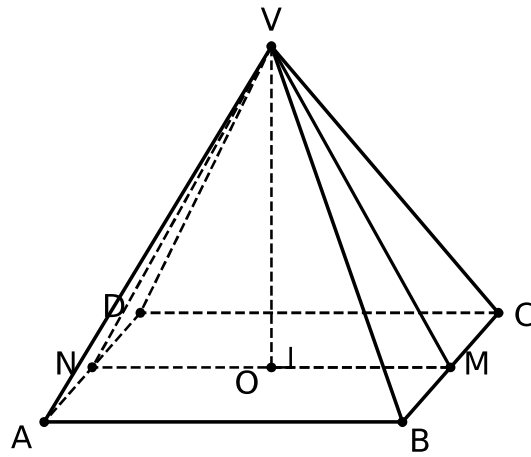


Figure 3

- (a) Figure 3 shows a right pyramid VABCD with a square base ABCD of side 16 **cm** . The height of the pyramid is 15 **cm** . Let M be the mid-point of BC and O be the centre of ABCD . Find VM .
- (b) Find the angle between the plane VBC and the base ABCD .
- (c) Let N be the mid-point of AD . A student claims that the angle between the plane VBC and the plane VAD is less than 60° . Do you agree? Explain your answer.

(8 marks)

19. (a) The equation of the circle C is $x^2 + y^2 - 10x - 24y = 0$. Denote the centre of C by G . Find the coordinates of G and the radius of C .
- (b) Let $L : 5x + 12y - 338 = 0$. Prove that L is a tangent to C , and find the coordinates of the point of contact T .
- (c) Denote the origin by O . Someone claims that OT is a diameter of C . Do you agree? Explain your answer.
- (d) Let P be a moving point on C such that P is different from O and T . Denote the area of $\triangle OPT$ by S . Find the greatest value of S .

(9 marks)

END OF PAPER